In [1]:
```
# file1.txt
# Hello World..!!!
```

Opening and reading from a file

In [3]:
```
fp = open("file1.txt","r")
data = fp.read()
print(data)
print(type(data))
fp.close()
```

```
Hello World..!!!
<class 'str'>
```

In [5]:
```
fp.name
```

Out[5]:
```
'file1.txt'
```

In [7]:
```
fp.mode
```

Out[7]:
```
'r'
```

In [9]:
```
fp.closed
```

Out[9]:
```
True
```

In [83]:
```
fp = open("file2.txt")
for i in fp:
    print(i, end="")
fp.close()
```

```
This is Python Programming.
This is second input file.
We are learning File handling.
This will be our future reference file.
```

In [15]:
```
fp = open("file.txt","r") #file.txt is not available
```

```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
Cell In[15], line 1
----> 1 fp = open("file.txt","r")

File ~\anaconda3\Lib\site-packages\IPython\core\interactiveshell.py:324, in _modi
fied_open(file, *args, **kwargs)
    317 if file in {0, 1, 2}:
    318     raise ValueError(
    319         f"IPython won't let you open fd={file} by default "
    320         "as it is likely to crash IPython. If you know what you are doin
g, "
    321         "you can use builtins' open."
    322     )
--> 324 return io_open(file, *args, **kwargs)

FileNotFoundError: [Errno 2] No such file or directory: 'file.txt'
```

In [17]:
```
fp = open("file1.txt") #default is r mode
print(type(fp))
fp.read(5) #can specify the number of characters want to read
fp.close()
```

```
<class '_io.TextIOWrapper'>
```

In [19]:
```
# file2.txt
# This is Python Programming.
# This is second input file.
# We are learning File handling.
# This will be our future reference file.
```

In [21]:
```
fp = open("file2.txt","r")
d = fp.readline() #read one line at a time
d1 = fp.readline()
print(d,end = "")
print(d1,end = "")
fp.close()
```

```
This is Python Programming.
This is second input file.
```

In [23]:
```
fp = open("file2.txt")
l = fp.readlines() #reads whole file and returns a list of all the lines
print(l)
fp.close()
```

```
['This is Python Programming.\n', 'This is second input file.\n', 'We are learnin
g File handling.\n', 'This will be our future reference file.']
```

In [25]:
```
fp = open("C:\\Users\\mania\\OneDrive\\Desktop\\file_desk.txt")
# absolute path to open a file from another directory
print(fp.read())
fp.close()
```

```
This file is coming from desktop.
```

error handling using with keyword

In [27]:
```
fp = open("C:\\Users\\mania\\OneDrive\\Desktop\\file_desk.txt")
# absolute path to open a file from another directory
print(fp.read())
print(10/0)
fp.close()
```

```
This file is coming from desktop.
---------------------------------------------------------------------------
ZeroDivisionError                         Traceback (most recent call last)
Cell In[27], line 4
      2 # absolute path to open a file from another directory
      3 print(fp.read())
----> 4 print(10/0)
      5 fp.close()

ZeroDivisionError: division by zero
```

In [29]:
```
fp.closed
```

Out[29]: False

In [31]:
```
with open("file2.txt","r") as fp:
    print(fp.read())
    print(10/0)
    fp.close()
```

```
This is Python Programming.
This is second input file.
We are learning File handling.
This will be our future reference file.
---------------------------------------------------------------------------
ZeroDivisionError                         Traceback (most recent call last)
Cell In[31], line 3
      1 with open("file2.txt","r") as fp:
      2     print(fp.read())
----> 3     print(10/0)
      4     fp.close()

ZeroDivisionError: division by zero
```

In [33]:
```
fp.closed
```

Out[33]: True

writing to a file

In [36]:
```
fp = open("file2.txt","w")
s1 = "Hello World."
fp.write(s1)
fp.close()
```

In [38]:
```
l1 = ["python\n","PC\n","SE\n","DBMS-II\n","Backend"]
```

In [40]:
```
fp = open("file2.txt","w")
fp.writelines(l1)
fp.close()
```

In [42]:
```
with open("file2.txt","r") as fp:
    print(fp.read().lower().count("file"))
    fp.close()
```

```
3
```

Sequential File Access

In [45]:
```
fp = open("abc.txt")
fp.read(2)
print(fp.read())
fp.close()
```

```
llo Students.
```

Random File Access

tell()

In [49]:
```
fp = open("abc.txt","r")
print(fp.tell()) #0
fp.read(3)
print(fp.tell()) #3
fp.read()
print(fp.tell()) #15
fp.close()
```

```
0
3
19
```

seek(offset, from_where = 0)

from_where:

0 : from the beginning of the file

1 : from the current position of the file pointer

2 : from the end of the file

In [54]:
```
# abc.txt
# Hello Students.
```

In [56]:
```
fp = open("abc.txt","r")
fp.read(5)
print(fp.read())
fp.close()
```

```
Students.
```

In [58]:
```
fp = open("abc.txt","r")
fp.read(5)
fp.seek(0,0) #defual from_where : 0
print(fp.read())
fp.close()
```

```
Hello Students.
```

In [62]:
```
fp = open("abc.txt","r")
fp.read(5)
fp.seek(2,0) #defual from_where : 0
print(fp.read())
fp.close()
```

```
llo Students.
```

In [64]:
```
fp = open("abc.txt","r")
print(fp.read(5))
fp.seek(0,1)
fp.close()
```

```
Hello
```

In [66]:
```
fp = open("abc.txt","r")
print(fp.read(5))
fp.seek(3,1)
fp.close()
```

```
Hello
```

```
---------------------------------------------------------------------------
UnsupportedOperation                      Traceback (most recent call last)
Cell In[66], line 3
      1 fp = open("abc.txt","r")
      2 print(fp.read(5))
----> 3 fp.seek(3,1)
      4 fp.close()

UnsupportedOperation: can't do nonzero cur-relative seeks
```

In [70]:
```
fp = open("abc.txt","rb")
print(fp.read(5).decode())
fp.seek(3,1)
fp.close()
```

```
Hello
```

In [72]:
```
fp = open("abc.txt","r")
print(fp.read(5))
fp.seek(0,2)
fp.close()
```

```
Hello
```

In [74]:
```
fp = open("abc.txt","rb")
fp.seek(-6,2)
print(fp.read().decode())
fp.close()
```

```
s.
```

File Access Modes

r, w, a, x, r+, w+, a+, x+ (for Text Files)

rb, wb, ab, xb, rb+, wb+ , ab+, xb+ (for Binary Files)

r :

reading only, if file exists then open for reading, file pointer is set at the beginning of the file and if the file does not exist then raise FileNotFoundError

w :

writing only, if file exists then open for writing by overwriting the old content, file pointer is set at the beginning of the file and if file does not exist then creates it.

a :

appending only, if file exists then open for writing at the end of the file, file pointer is set at the end of the file and if file does not exist then creates it.

x :

exclusive creation (writing only), if file exists then raise FileExistsError and if file does not exist then creates a new file and file pointer is set at the beginning of the file.

r+ :

reading and writing, if file exists then open for both reading and writing, file pointer is set at the beginning of the file and if file does not exist then raise FileNotFoundError.

w+ :

reading and writing, if file exists then open by overwriting the old content, file pointer is set at the beginning of the file and if file does not exist then creates it.

a+ :

reading and appending, if file exists then open for reading and writing, file pointer is set at the end of the file for writing and if file does not exist then creates it.

x+ :

exclusive creation (reading and writing), if file exists then raise FileExistsError and if file does not exist then creates a new file and file pointer is set at the beginning of the file.